

The Design Process for Google's Training Chips: TPUv2 and TPUv3

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Five years ago, few would have predicted that a software company like Google would build its own chips. Nevertheless, Google has been deploying chips for machine learning (ML) training since 2017, powering key Google services. These tensor processing units (TPUs) are composed of chips, systems, and software, all codesigned in-house. This article details the circumstances that led to this outcome, the challenges and opportunities observed, the approach taken for the chips, a quick review of performance, and finally a retrospective on the results. A companion paper describes the supercomputers built from these chips, the compiler, and its performance in detail.⁶

FORCES PUSHING TOWARDS CUSTOM ML HARDWARE

In 2013, only one year after AlexNet swept the ImageNet competition,⁴ Google leaders predicted ML would soon be ready to attack difficult problems like production versions of image and speech recognition. Alas, these ML models were so computationally expensive that a sustainable service was nearly impossible, as Internet service costs scale by the number of users who grow as the service improves. The motivation to slash ML inference serving costs led to the TPUv1 inference system, deployed successfully in 2015.⁵

TPUv1 exposed the next bottleneck: ML training. The team quickly pivoted into building the TPUv2 training system. Two years later, TPUv2 powered key Google services with fast and cost-effective training.

CHALLENGES AND OPPORTUNITIES OF BUILDING ML HARDWARE

ML training brings challenges relative to ML inference:

- *More computation.* More means both the types of computation—for example, backpropagation

requires matrix transposition and loss functions—and the *amount* of computation. An inference calculation typically executes on the order of 10^9 floating point operations, but Google's production training applications require 10^{19} – 10^{22} ; more than 10 orders of magnitude larger!

- *More memory.* During training, temporary data are kept longer for use during backpropagation. With inference, there is no backpropagation, so data are more ephemeral.
- *Wider operands.* Inference can tolerate int8 numerics relatively easily, but training is more sensitive to dynamic range due to the accumulation of small gradients during weight update.
- *More programmability.* Much of training is experimentation, meaning unstable workload targets such as new model architectures or optimizers. The operating mode for handling a long tail of training workloads can be quite different from a heavily optimized inference system.
- *Harder parallelization.* For inference, one chip can hit most latency targets. Beyond that, chips can be scaled out for greater throughput. In contrast, exaflops-scale training runs need to produce a single, consistent set of parameters across the full system, which is easily bottlenecked by off-chip communication.

These problems felt daunting initially, plus we had constraints on time and staffing. Time matters because each day saved during development accelerates our production training pipeline a day. And as for staffing: while Google is teeming with engineers, they are not all available for our project. Ultimately, the TPU team had only a cup from Google's large engineering pool. Thus, we had to be ambitious to overcome the complexities of training, but the time and staffing budget set constraints.

To prioritize, we sorted activities into two buckets: those where we must do a great job, and those that we only have to do good enough. The first bucket included the following:

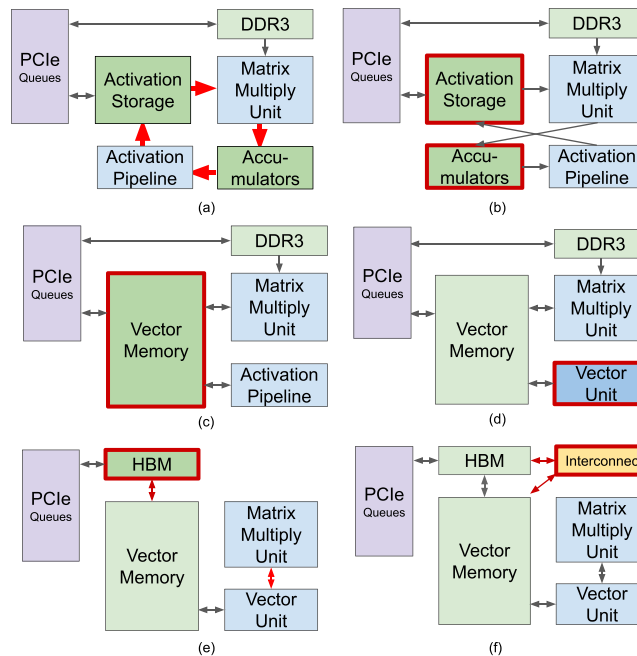


FIGURE 1. Transforming the TPUv1 datapath into the TPUv2 datapath.

- ① Build quickly.
- ② Achieve high chip performance.
- ③ Scale efficiently to numerous chips.
- ④ Work for new workloads out of the box.
- ⑤ Be cost effective.

Everything else was in the second bucket. While tempting to brush these second bucket issues aside as minor embarrassments, the reality is building and delivering a good system is as much about what you decide *not* to do as what you decide to do. In retrospect, these decisions are not embarrassing after all!

We refer to the relevant goals using the circled numbers (e.g., ②) throughout the discussion to highlight “first bucket” design decisions.

OUR APPROACH TO ML HARDWARE

TPUv1 provided a familiar starting point for our training chip [see Figure 1(a)]. The high-bandwidth loop (red) identifies the important part of TPUv1: the core data and computation loop that crunches neural network layers quickly. DDR3 DRAM feeds the loop at much lower bandwidth with model parameters. The PCIe connection to the host CPU exchanges model inputs and outputs at even lower bandwidth.

Figure 1 shows five piecewise edits that turn TPUv1 into a training chip. First, splitting on-chip SRAM makes sense when buffering data between sequential

fixed-function units, but that is bad for training as on-chip memory requires more flexibility. The first edit merges these into a single vector memory [see Figure 1 (b) and (c)]. For the activation pipeline, we moved away from the fixed-function datapath (containing pooling units or hard-coded activation functions) and built a more programmable vector unit [see Figure 1 (d)] ④. The matrix multiply unit (MXU) attaches to the vector unit as coprocessor [Figure 1(e)].

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Loading the read-only parameters into the MXU for inference does not work for training. Training writes those parameters, and it needs significant buffer space for temporary per-step variables. Hence, DDR3 moves behind Vector Memory so that the pair form a memory hierarchy [also in Figure 1(e)]. Adopting in-

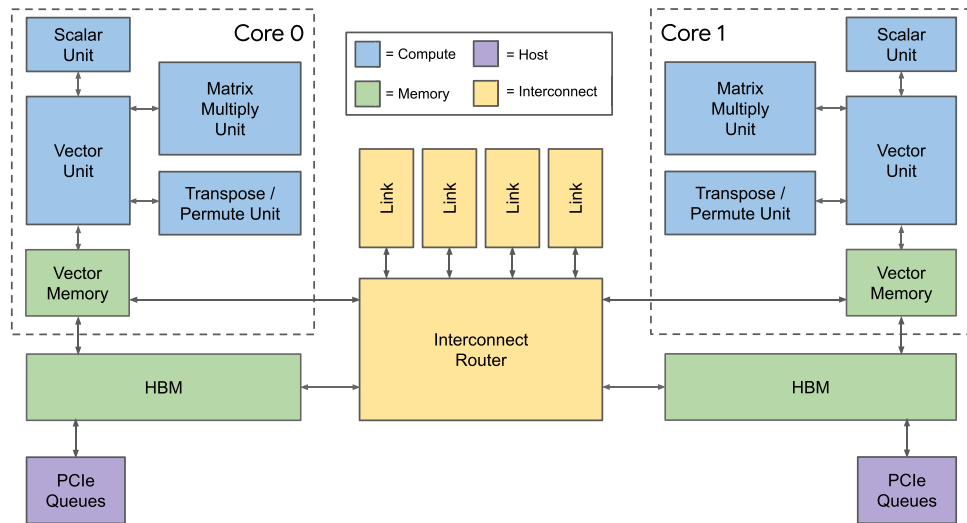


FIGURE 2. TPUv2 datapath in more detail, showing both cores, which appear as a single PCIe device.

package HBM DRAM instead of DDR3 upgrades bandwidth twentyfold, critical to utilizing the core ②.

Last is scale. These humongous computations are much bigger than any one chip. We connect the memory system to a custom interconnect fabric (ICI for InterChip Interconnect) for multichip training [see Figure 1(f)] ③. And with that final edit, we have a training chip!

Figure 2 provides a cleaner diagram, showing the two-core configuration. The TPUv2 core datapath is blue, HBM is green, host connectivity is purple, and the interconnect router and links are yellow. The TPUv2 Core contains the building blocks of linear algebra: scalar, vector, and matrix computation.

Why two cores? The simpler answer is that we could fit a second core into a reasonably sized chip ②⑤. But then why not build a single, bigger core? Wire latencies grow as the core gets bigger, and the two-core configuration hits the right balance between reasonable pipeline latency and additional per-chip computation capacity. We stopped at two bigger cores because they are easier to program as they allow the developer to reason about one unified instruction stream operating on big chunks of data ④, rather than exposing lots of tiny cores that need to be lashed together. We took advantage of training fundamentally being a big-data problem.

The following sections dive deeper into key chip components.

TPUV2 CORE

The TPUv2 Core was codesigned closely with the compiler team to ensure programmability ④. (The

compiler's name is XLA, for accelerated linear algebra.⁶) In these discussions, we landed on two important themes. First, a VLIW architecture was the simplest way for the hardware to express instruction level parallelism and allowed us to utilize known compiler techniques ①②. Second, we could ensure generality by architecting within the principled language of linear algebra ④. That meant focusing on the computational primitives for scalar, vector, and matrix data types.

Scalar Computation Unit

The scalar unit is where computation originates. It fetches complete VLIW bundles from a local instruction memory, executes the scalar operations slots locally, and then forwards decoded instructions on to the vector and matrix units where execution happens later, decoupled from scalar execution. The VLIW bundle is 322 bits and is composed of two scalar slots, four vector slots (two used for vector load/store), two matrix slots (a push and a pop), one miscellaneous slot (a simple example would be a delay instruction), and six immediates.

Figure 3(a) shows a diagram of the scalar unit. At the top left is the instruction bundle memory. While an instruction cache backed by HBM would have been nice, a DMA target for software-managed instruction overlays was easier ①. It is not the flashiest solution, but remember that this part just needs to be "good enough." The greatness in this system lies elsewhere.

Down the left of Figure 3(a), we perform scalar decode and issue, and on the right is where scalar execution happens. At the bottom right is the DMA port into

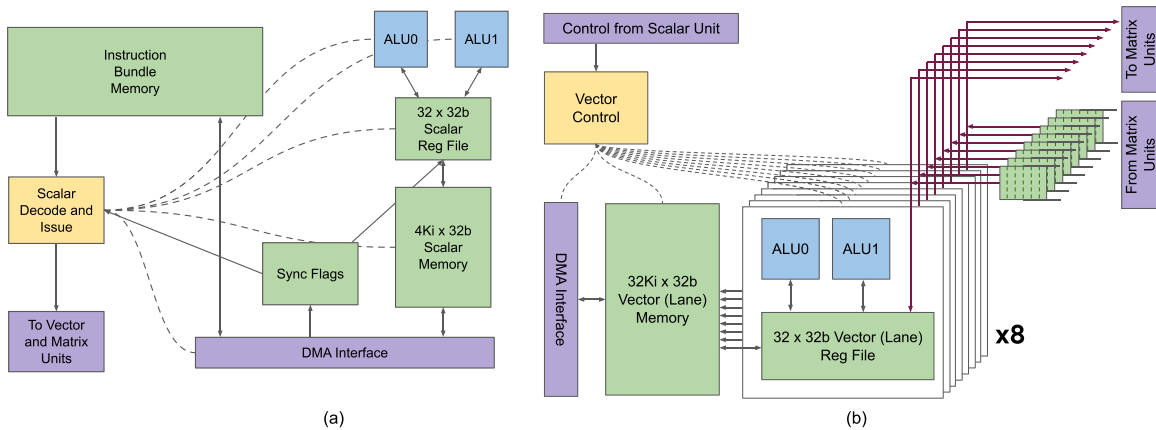


FIGURE 3. (a) TPUv2 core scalar unit and (b) single vector lane. The vector unit contains 128 vector lanes.

the memory system, primarily to HBM. That feeds into local Scalar Memory SRAM to issue loads and stores against. They feed a 32-deep register file containing 32-bit words, which finally feeds into a dual-issue ALU at the top right. Execution interlock is managed by standard hold conditions on the instructions, while synchronization flags provide a way to interlock against software-managed DMA operations. The memory hierarchy is under the control of the compiler, which simplifies the hardware design while delivering high performance ①②③.

Vector Computation Unit

After scalar execution, the instruction bundle and up to three scalar register values are forwarded to the vector unit. Figure 3(b) shows a single vector lane, and the entire vector unit contains 128 such vector lanes. Each lane contains an additional 8-way execution dimension called the *sublane*. Each sublane contains a dual issue 32-bit ALU connected into a 32 deep register file. All together, the vector computation unit allows operation on eight sets of 128-wide vectors per clock cycle. Sublanes let TPUv2 increase the vector versus matrix compute ratio, which is useful for batch normalization.⁶

Each lane's register files perform loads and stores against the lane's local slice of vector memory, and that memory connects into the DMA system (primarily providing access to HBM).

On the right of Figure 3(b) is the connectivity into the matrix units. The push instruction slots send data vectors out to the matrix units. A *result FIFO* captures any returning result vectors from the matrix units, and these can be popped into vector memory using the pop instruction slots. The result FIFO lets us avoid strict execution schedule constraints for the long-latency matrix operations and shorten register lifetimes, simplifying the compiler ①.

Matrix Computation Units

The matrix multiply unit is the computational heart of the TPU. It is a 128 x 128 systolic array of multipliers and adders, delivering 32,768 operations per cycle ②. The inputs are two matrices: a left hand side matrix and a right hand side matrix. The left hand side matrix streams over a preloaded right hand side matrix to create a streaming result matrix, which is sent directly to the vector unit's result FIFO. The right hand side matrix can be optionally transposed when loaded into the matrix unit.

Critically, the systolic array structure provides high computational density ②. While it performs most of the FLOPS per second, it is not the largest contributor to chip area (see Figure 4) ⑤.

Numerics shape computation density. Multiplications use *bfloat16*: this is a 16-bit format that has the same exponent range as float32 but with fewer bits of mantissa. The accumulation happens in full 32-bit floating point.

Bfloat16 works seamlessly for almost all ML training, while reducing hardware and energy costs ②⑤. Our estimate for the more recent 7 nm is that bfloat16 has a $\approx 1.5\times$ energy advantage over the IEEE 16-bit float: 0.11 versus 0.16 pJ for add and 0.21 versus 0.31 pJ for multiply. Moreover, bfloat16 is easier for ML software to use than fp16, since developers need to perform loss scaling to get fp16 to work.⁹ Many are willing to do the extra programming to go faster, but some are not. For example, all but 1 of the 200 ML experts at the Vector Institute rely on the slower 32-bit floating point in GPUs.⁸ We know of no downsides for bfloat16 versus fp16 for ML. As it takes less area and energy and is easier for ML software to use ②④, bfloat16 is catching on. TPUv2 was the first, but ARM, Intel, and NVIDIA have subsequently embraced bfloat16.

Matrix multiplication is not the only important matrix transformation, so another set of units efficiently perform other matrix operations like transposes, row reductions, or column permutations.

TPUV2 MEMORY SYSTEM

As discussed earlier, the TPUv2 Core has a variety of SRAM-based scratchpad memories. These are architecturally visible to software and accessed using loads and stores. This approach gives the compiler predictable execution control over both computation and SRAM and simplifies the hardware ②④.

But the memory story cannot stop with SRAM, because most models would not fit. Figure 2 shows that high-bandwidth, in-package HBM backs up the SRAM. The compiler moves data between HBM and SRAM using asynchronous DMAs. Since HBM holds vectors and matrices, DMAs can stride through memory to allow slicing off subdimensions of larger dimensional structures to reduce the DMA descriptor overhead. When a DMA finishes, a completion notification lands in the core's sync flags, allowing the program to stall until data arrives.

Taking a step back, this approach provides a clean partitioning of concerns. The core (blue in Figure 2) provides predictable, tightly scheduled execution ②. The memory system (in green) handles asynchronous prefetch DMA execution from the larger HBM memory space ⑤. The hand-off between the regimes is managed with sync flags.

Ultimately, the primary goal of the memory system is to keep the datapath fed ②, so HBM's high bandwidth is critical. At 700 GB/s per chip, the HBM of TPUv2 provides 20× the bandwidth of the pair of DDR3 channels in TPUv1. This increase allows full computation utilization at much lower data reuse rates ④.

Zooming out to the chip-level memory system, Figure 2 shows each core is connected to half of the chip's HBM. The split HBM memory space might be a bit surprising, but the reason is simple: we wanted to build this chip quickly ①, this approach simplified the design, and it was good enough.

Each core also has a set of PCIe queues to exchange data with the host. Between the two cores is the interconnect router that also connects to the off-chip links.

TPUV2 INTERCONNECT

A dedicated interconnect is the foundation of the TPUv2 supercomputer ("pod"). TPUv1 was a single-chip system built as a coprocessor, which works for inference. Training Google production models would require months on a single chip. Hence, TPUv2s can

connect into a supercomputer, with many chips working together to train a model.

The chip includes four off-chip links and two on-chip links connected with an on-chip router. These four links enable the 2-D torus system interconnect, which matches many common ML communication patterns, like all-reduce. The four external links deliver 500 Gb/s and the two internal links are twice as fast. Some torus links are within the TPU tray, and the rest are through cables across trays and racks.

An important property of interconnect design is ease of use ④; DMAs to other chips look just like DMAs to local HBM, albeit with a push-only restriction for simplicity ①. This common DMA interface allows the compiler to treat the full system's memory space consistently.

The dedicated TPU network enables scalable synchronous training across TPUs. Asynchronous training was the state-of-the-art previously, but our studies showed synchronous converges better ④—async primarily allowed broader scaling when networking was limited.

We connect the TPU pod to storage over the data-center network to feed the input data for the model via a PCIe-connected CPU host. The system balance across CPU, network, and storage is critical to achieve end-to-end performance at scale ③. The PCIe straw is tiny (16 GB/s per chip) the in-package bandwidth is huge (700 GB/s per chip), and the interconnect links are somewhere in the middle (4× 60 GB/s). Building TPUv2 to be flexible and programmable allows operation on all data locally instead of moving data back to the host through a bandwidth-constricted PCIe bus ②.

TPUV2 FLOOR PLAN

The floorplan in Figure 4 is not stylish, but it was good enough and allowed us to focus on more important things ①. Most area is for the computation cores in blue, with noticeable area for the memory system and interconnect. The two cores are split across the top and bottom, with the interconnect router sitting in the donut hole in the middle. The white areas are not empty, but filled with wiring.

The two matrix multiply units are at the top center and bottom center. The FLOP/second delivered in such a small area demonstrates the computation density available with the bfloat16 systolic array ⑤.

TPUV3

We hoped to avoid the second system effect²; we did not want to blow everything we worked hard for in TPUv2 by building the kitchen sink into TPUv3. TPUv3 is a "mid-life

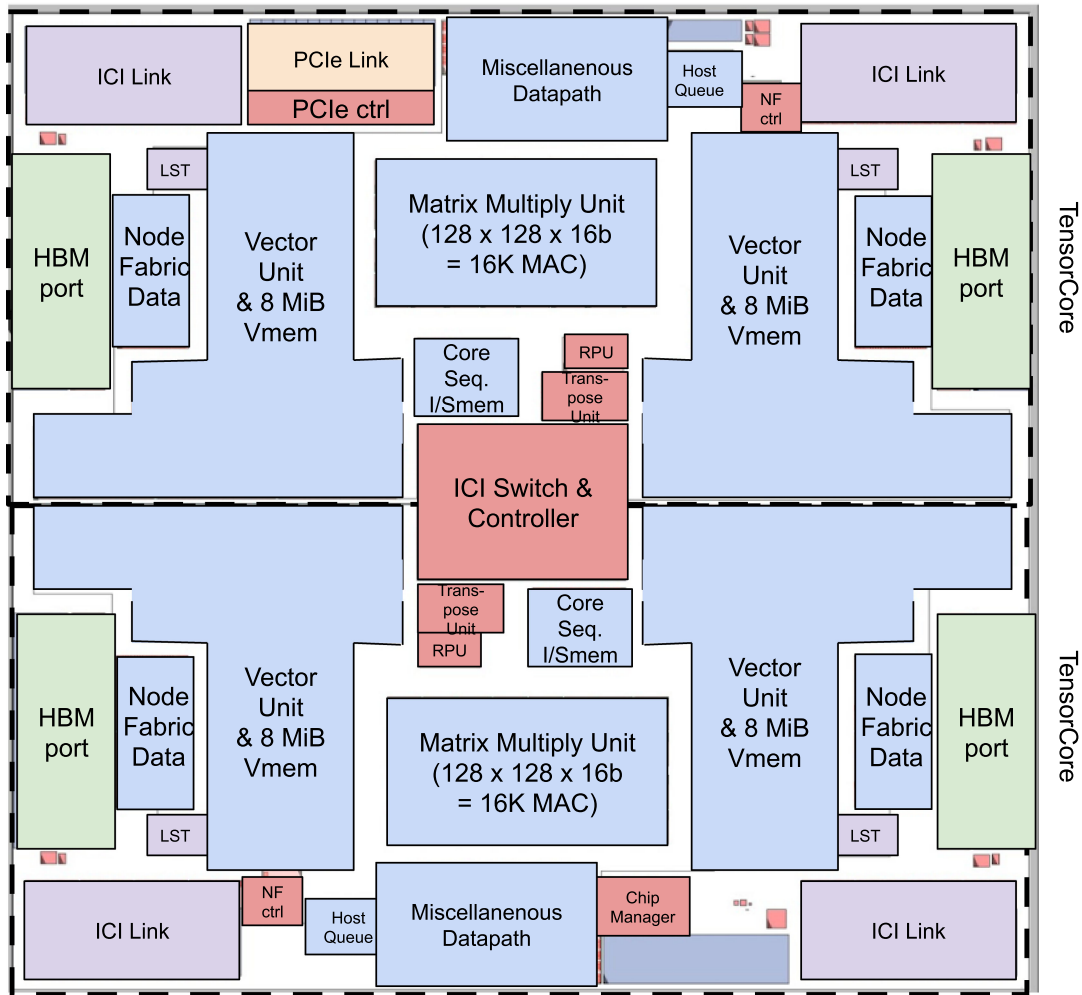


FIGURE 4. TPUv2 floor plan.

kicker” that leveraged what we already built—both use 16 nm technology—but let us pick the low-hanging fruit left behind by the quick development of TPUv2 ①. The most important enhancements were the following.

- Doubling the matrix multiply units to get double maximum FLOPS/second ② (high leverage due to the matrix multiply unit computation density).
- Increasing clock frequency from 700 to 940 MHz ② (30% performance from tuning pipelines).
- Bumping HBM performance by 30% using a higher bus speed ②.
- Doubling HBM capacity enabling bigger models and bigger batch sizes [see Figure 5(a)] ④.
- Raising interconnect link bandwidth by 30% to 650 Gb/s per link ②.
- Raising the maximum scale to a 1024-chip system, up from the 256-chip limit of TPUv2 ③.

It seems quaint now, but we thought the 256-chip TPUv2 system was huge. ML’s voracious appetite continues,¹ so moving to 1024-chip systems was critical ③.

PERFORMANCE SYNOPSIS

Figure 5 shows a roofline model of TPUv2/v3 and the NVIDIA V100 GPU. It identifies programs as memory-bound (under the slanted roofline) or compute-bound (under the flat roofline) using *operational intensity* (operations per DRAM byte accessed) to place applications appropriately. The vertical lines in Figure 5(a) show the performance improvements from TPUv2 to TPUv3. CNN0 is actually above its fp32 roofline in Figure 5(b)! This model includes many 3×3 convolutions and V100 uses Winograd Transform that does fewer operations than a

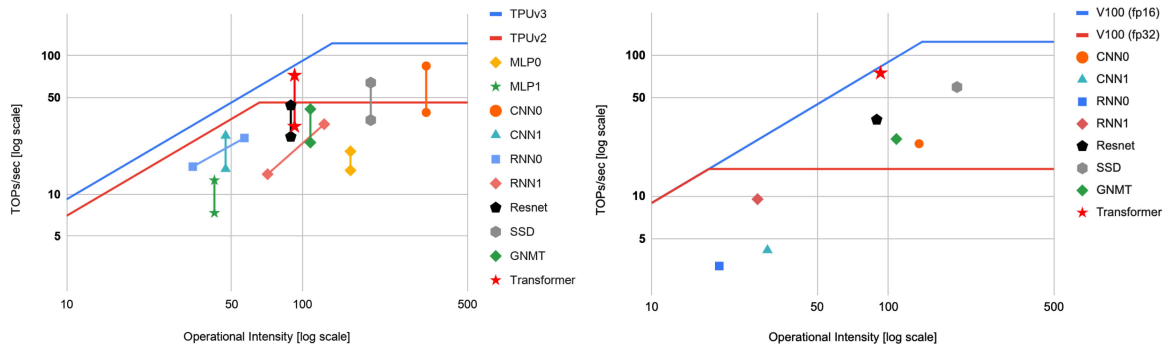


FIGURE 5. (a) Roofline Models¹¹ for TPUv2 and TPUv3 (left). (b) Roofline Models of V100 for fp16 and fp32 (right). RNN0/RNN1 move up and to the right in (a) since the larger HBM capacity of TPUv3 enabled bigger batch sizes; all others use the same batch size. MLP0/MLP1 could not run on Volta in (b) because the embedding tables were too big.

convolution. However, Figure 5(b) assumes the same computation for comparison, lifting CNN0 through its roofline.

AS DEEP LEARNING CONTINUES TO EVOLVE, AND AS WE UNDERSTAND THE WORKLOADS AND SCALING LIMITS BETTER, OPPORTUNITIES CONTINUE FOR FURTHER CODESIGN ACROSS ML MODELS, SOFTWARE, AND HARDWARE TO IMPROVE NEXT GENERATION TPUS AND OTHER DOMAIN-SPECIFIC ARCHITECTURES.

A companion paper⁶ measures performance in detail, summarized here.*

- Although TPUv3 is a smaller chip (<700 versus 812 mm²) in an older semiconductor technology than Volta (16 nm versus 12 nm), the geometric mean of performance per chip is the same for the MLPerf 0.6 benchmarks⁵. GPUs are larger in part because they also perform graphics operations.
- Using Google production applications, TPUv3 is on average five times faster. Like the Vector Institute, Googlers use 32-bit floating point operations on V100 [see Figure 5(b)] as they were

unwilling to do the extra work of loss scaling for fp16⁹ ④.

- The peak computation gain of TPUv3 over TPUv2 is 2.7×, but the improvements in memory bandwidth, ICI bandwidth, and clock rate are only 1.3×. Were the extra MXUs in TPUv3 underutilized due to bottlenecks elsewhere? The geometric mean of performance gains of TPUv3 over TPUv2 were 1.8× for both MLPerf 0.6 benchmarks and Google production applications². MXUs are a small part of the chip (see Figure 4), so doubling the MXUs worked well⁵.
- Google applications scaled to 96%–99% of perfect linear speedup³; applications for general-purpose supercomputers are rarely that high.
- ③ allows TPUv3s to scale to supercomputer size and performance. The GigaFLOPS/Watt of a TPUv3 supercomputer running a Google application was 50× higher than a general-purpose supercomputer running the Linpack benchmark⁶ ⑤.

CONCLUSION

TPUv2/v3 met our “first bucket” goals:

① Build quickly.

Our cross-team co-design philosophy found simpler-to-design hardware solutions that also gave more predictable software control, such as DMA to main memory (HBM) and compiler-controlled on-chip memory instead of caches. Along the way, we made difficult tradeoffs to preserve the development schedule, such as splitting the HBM between the two cores, tolerating an inefficient

*MLPerf 0.7 previewed the next generation TPU chip (TPUv4), which matched the performance of NVIDIA's next generation GPU chip (Ampere), but that's a story for another time.

chip layout, and using FIFOs to simplify XLA compiler scheduling.

② *Achieve high performance.*

Matrix computation density, high HBM bandwidth, and XLA compiler optimizations deliver excellent performance.

③ *Scale up.*

Our system-first approach and simple-to-use interconnect let TPUv3 scale natively to 1024 chips and deliver nearly linear speedup for production applications.

④ *Work for new workloads out of the box.*

To support the deluge of training workloads, we built a core grounded in linear algebra that works well with the XLA compiler, and HBM ensures we have enough capacity and bandwidth to keep pace with growing models.

⑤ *Be cost effective.*

The matrix units are efficient, the design was simple without gratuitous bells and whistles, and we got our money's worth in performance.

As deep learning continues to evolve, and as we understand the workloads and scaling limits better, opportunities continue for further codesign across ML models, software, and hardware to improve next generation TPUs and other domain-specific architectures.

ACKNOWLEDGMENTS

TPUv2/v3 exist thanks to the creativity and hard work of the TPU team, from chips to systems to software. The author list represents a small slice, and we are grateful for the opportunity to work together. Thanks go to Paul Barham, Eli Bendersky, Dehao Chen, Clifford Chao, Chiachen Chou, Jeff Dean, Brian Fosco, Ben Gelb, Jesse Guss, Peter Hawkins, Blake Hechtman, Mark Heffernan, Richard Ho, Robert Hundt, Michael Isard, Terry Kang, Fritz Kruger, Naveen Kumar, Sameer Kumar, Steve Lacy, Chris Leary, Hyoun-Joong Lee, David Majnemer, Lifeng Nai, Rahul Nagarajan, Tayo

Oguntebi, Andy Phelps, Paul Rodman, Bjarke Roune, Brennan Saeta, Amir Salek, Julian Schrittwieser, Dan Steinberg, Andy Swing, Horia Toma, Shibo Wang, Tao Wang, Yujing Zhang, and many more.

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